

Ideation Phase - Brainstorm & Idea Prioritization Report

Automated Network Request Management in ServiceNow

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Team ID	Individual / Solo Project
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Project Title	Automated Network Request Management in ServiceNow

1. Project Overview

The Automated Network Request Management project in ServiceNow streamlines how employees request, get approval for, and receive network configuration changes (such as firewall rules, port openings, and VPN access) across the organization. Traditionally, such requests travel through emails or verbal communication with IT staff, which creates inconsistent approval trails, delayed fulfillment, poor visibility into request status, and security compliance risks. This project replaces that ad-hoc process with a standardized, auditable Service Catalog item backed by automated workflows spanning approval, security validation, and deployment.

By digitizing the request lifecycle end-to-end, the organization gains a single system of record for every network request, enforces consistent approval and InfoSec policies, and reduces the manual effort network teams spend tracking configurations and security audits.

2. Step 1: Team Gathering & Problem Statement

2.1 Problem Statement

Network access and configuration requests are commonly processed through emails or informal channels, causing inconsistent approvals, delayed fulfillment, lack of request visibility, and critical security compliance issues. This project proposes a standardized ServiceNow Service Catalog solution that automates request submission, manager/security approvals, risk verification, task creation, and network fulfillment.

2.2 How Might We?

How might we design a standardized ServiceNow Service Catalog workflow that automates network approvals, security validation, and fulfillment while eliminating manual delays and compliance risks?

2.3 Session Goals

- Replace manual, email-based requests with a structured Service Catalog item.
- Automate approvals using Flow Designer, including manager and InfoSec risk-based approval paths.
- Integrate the catalog workflow with the CMDB/Network Inventory for accurate topology tracking.
- Enable automated deployment through IntegrationHub or network automation tools (e.g., Ansible).
- Improve transparency using request tracking, notifications, and self-service status visibility.

3. Step 2: Idea Listing & Grouping

Ideas generated during the ideation session were grouped into three functional areas covering the catalog experience, the underlying workflow logic, and the fulfillment mechanics.

3.1 Group A: Service Catalog

- **A1-Dynamic Catalog Item:** Custom-built catalog item that dynamically shows network options (VPN, Port, Firewall) based on department, role, or access requirements.
- **A2-Real-time Policy Indicator:** Real-time indicator on the catalog item flagging whether a requested port/protocol violates baseline security policies before submission.
- **A3-Self-Service Status Tracker:** A self-service portal widget where employees can track the live status of their network request without contacting IT.

3.2 Group B: Workflow

- **B1-Tiered Security Approval:** Multi-level manager and InfoSec approval chain that scales based on the requested access level or risk tier.
- **B2-Timeout Escalation:** Automatic escalation to the next approver or IT admin if an approval is not actioned within a defined SLA window.
- **B3-Dynamic Risk Assessment:** Flow logic that evaluates the requested network change against a risk catalog and routes high-risk configuration changes through an additional security review board.

3.3 Group C: Fulfillment

- **C1-CMDB Integration:** Direct integration with the Configuration Management Database to update network topology and CI relationships automatically on approval.
- **C2-Ansible/IntegrationHub Deployment:** Integration Hub spokes connecting to network automation platforms (like Ansible) to push zero-touch firewall/port configurations to network devices.
- **C3-Automated Ticket Closure:** Automatic closure of the Requested Item (RITM) and parent Request (REQ) once the fulfillment task confirms successful network deployment.

4. Step 3: Idea Prioritization

Each idea was plotted against Importance (business/user value) and Feasibility (ease of implementation within the current ServiceNow instance and available integrations) to decide what belongs in the Minimum Viable Product (MVP) versus later phases.

Priority Quadrant	Ideas
High Importance / High Feasibility	Dynamic Catalog Item, Tiered Manager/Security Approval, Automated RITM Closure
High Importance / Low Feasibility	CMDB Integration, Zero-Touch (Ansible/IntegrationHub) Deployment
Low Importance / High Feasibility	Self-Service Status Tracker, Timeout Escalation
Low Importance / Low Feasibility	Real-Time Security Policy Validator

MVP Scope: The High Importance / High Feasibility items form the core MVP — a dynamic catalog item, tiered manager/security approval, and automated ticket closure — since they deliver the greatest immediate reduction in manual effort with the least implementation risk. High-value but low-feasibility items such as CMDB integration and zero-touch deployment are targeted for Phase 2 once the core flow is stable, as they depend on external system connectivity (CMDB data model, REST APIs).

5. Proposed Solution Architecture

The end-to-end flow moves a request from submission to fulfillment without manual intervention, branching only where a security or approval decision is required:

Service Portal Catalog Item → Flow Designer Trigger → Risk Type Check → Standard Risk (Manager Approval) / High Risk (InfoSec Approval) → CMDB Validation → IntegrationHub Deployment (Ansible) → Network Configured → Request Item Automatically Closed.

5.1 Architecture Stage Breakdown

- **Service Portal Catalog Item:** The employee selects the desired network access from a Service Catalog item exposed on the Service Portal.
- **Flow Designer Trigger:** Submission of the Requested Item (RITM) fires a Flow Designer flow that governs the rest of the lifecycle.
- **Risk Type Check:** The flow inspects catalog variables to classify the request as standard/low-risk or high-risk.
- **Approval Branch:** Standard requests proceed directly to manager approval; high-risk requests are routed to both the manager and an InfoSec team member for sign-off.
- **CMDB Validation:** On approval, the system checks the Configuration Management Database to ensure the target device/network segment exists and is active.
- **IntegrationHub Deployment:** An IntegrationHub spoke calls an automation platform (e.g., Ansible) to deploy the network change to the target endpoint.
- **Automated Closure:** A confirmation callback marks the catalog task complete, closes the RITM and REQ, and notifies the employee via email/portal notification.

6. Expected Benefits

- Faster turnaround from request to network access through automated routing and deployment.
- Full audit trail of every approval, security verification, and configuration change for compliance reporting.
- Reduced IT workload by eliminating manual ticket updates and baseline configuration changes.
- Better network topology visibility through direct CMDB integration.
- Improved employee experience via self-service submission and status tracking.

7. Next Steps

- Build the Service Catalog item and variable set for network request selection in a Personal Developer Instance (PDI).

- Design the Flow Designer workflow covering manager/security approvals, CMDB validation, and fulfillment task creation.
- Configure Integration Hub connectivity for REST/Ansible (or simulate with a mock fulfillment task for demo purposes).
- Test end-to-end with both standard and high-risk scenarios, including approval timeout escalation.
- Document the solution and prepare a walkthrough demo for evaluation.